Do Now · retrieval

Magnetic, optical and flash storage

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why could a GPU be useful for many independent pixel operations?

Q2

What does ACC hold?

Q3

Define non-volatile.

Q4

Evaluate 11010110_2 AND 00001111_2, then write the resulting byte in hexadecimal.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify the storage mechanism — HDDs use magnetised regions on rotating platters; DVD-style media are read optically.

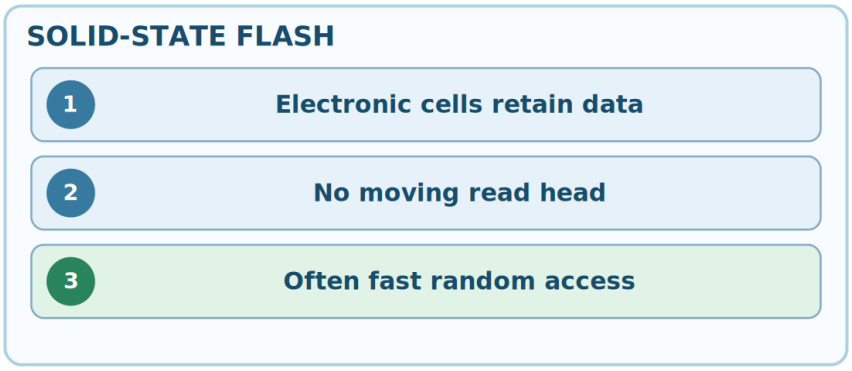
Core idea 2

Explain solid-state storage — Flash cells are non-volatile electronic storage without rotating platters or mechanical head seeks.

Core idea 3

Evaluate rather than rank universally — SSD generally has low seek latency and resists mechanical shock; HDD may offer lower cost per GB at large capacities.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

magnetic — Bits stored through magnetic states.

optical — Bits encoded/read through optical disc patterns.

flash — Non-volatile electronic storage.

seek latency — Delay in finding data.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

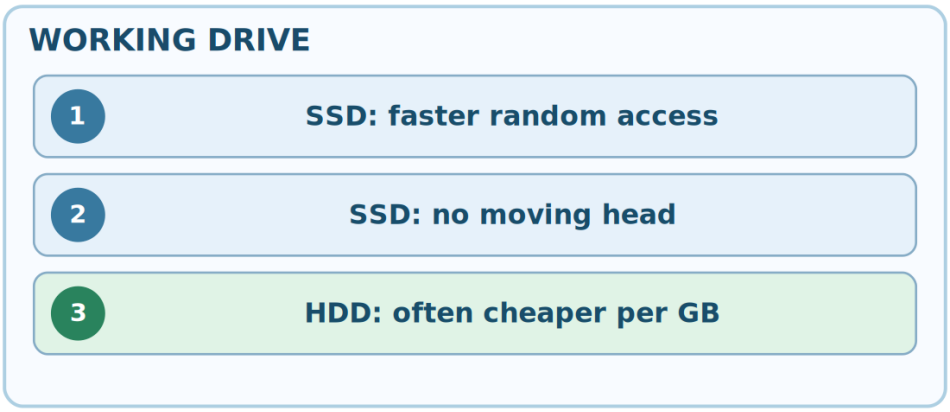
Problem

Choose SSD or HDD for a photographer moving large images between sites.

Follow the worked method

1. Predict — An SSD is attractive for frequent portable access and avoiding moving parts.
2. Explain — An HDD can offer capacity at a lower cost; backups remain necessary regardless.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

An optical disc can distribute static content without needing a network connection.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give storage technology and one physical feature for HDD, DVD and USB flash.

Question 2

Explain two benefits and one drawback of an SSD rather than an HDD for frequent portable editing.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Propose fast working storage and bulk archival storage for a school, explaining backup strategy.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

USB flash and HDDs use the same magnetic storage method.

Use key vocabulary from memory

1. Explain magnetic in your own words.

2. Explain optical in your own words.

3. Explain flash in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.